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RESEARCH PAPER

Experimental Framework and Multi-Response Analysis of Support-Assisted Annealing of FFF-Printed PLA for Improved Mechanical Performance and Dimensional Stability

Rapid Prototyping, Tooling and Manufacturing

Production and Industrial Engineering

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Abstract

Thermal annealing can modify the frozen molecular orientation, crystallization state, residual stress and inter-road interfaces of fused-filament-fabricated (FFF) polylactic acid (PLA). These changes may improve heat resistance or selected mechanical responses, yet the same increase in chain mobility can cause anisotropic shrinkage, thickness growth and warpage. Published studies have examined conventional oven treatment and support-assisted routes using sand, ceramic powder, salt, gypsum, liquids and elastomeric encapsulation, but their conclusions are difficult to compare because PLA formulation, geometry, print architecture, temperature regime and response definitions differ. This paper develops a feasible, standards-informed framework for a controlled comparison of as-printed, unconstrained, sand-supported, salt-supported and mechanically fixture-constrained PLA. A 25-specimen screen at 95 C for 60 min is followed by a replicated 3 temperature x 3 duration x 2 constraint-state factorial, tensile confirmation and limited flexural confirmation, for 130 physical specimens. Primary responses are ultimate tensile strength, three-axis dimensional change and warpage; modulus, strain at break, flexural response and optional differential scanning calorimetry provide supporting evidence. Block-aware analysis, interaction modelling, confidence intervals, Pareto comparison and Derringer-Suich desirability are specified to balance mechanical performance against dimensional error. Because student measurements are not yet available, the Results and Discussion section is a critical engineering synthesis of verified literature rather than a report of local test values. The framework provides a practical basis for determining whether low-cost support can preserve useful annealing effects while limiting geometric loss in tolerance-sensitive FFF components.

Keywords: additive manufacturing; fused filament fabrication; polylactic acid; annealing; dimensional stability; mechanical properties; design of experiments; multi-response optimization

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1. Introduction

Material extrusion deposits softened thermoplastic roads along programmed toolpaths. Successive roads must consolidate while cooling, so bead placement is a transient welding problem rather than simple geometric stacking. ISO/ASTM 52900 defines the terminology and ISO 17295 provides the orientation system [1], [2]. At interfaces, contact, wetting and molecular interdiffusion compete with cooling; at part scale, raster and build direction establish preferred load paths [9]-[17].

PLA is accessible and comparatively easy to print, but its thermal response depends on stereochemistry, molar mass, moisture, additives, pigment and prior cooling [18]-[21]. An FFF coupon is a mesostructured assembly of roads, interfaces, voids and frozen orientation, not homogeneous bulk PLA. Identical nominal annealing conditions can therefore produce different outcomes across grades and print strategies.

Annealing is attractive because it can be implemented with relatively simple equipment. Heating above the glass-transition region increases segmental mobility, enabling stress relaxation, further crystallization and structural rearrangement. Depending on formulation and thermal history, these mechanisms may increase useful stiffness, thermal stability or inter-road cohesion [28]-[37]. They also permit the part to move. Length, width and thickness may change in different directions; out-of-plane curvature may grow; and a mechanically stronger coupon may no longer meet a drawing tolerance. Mechanical improvement and dimensional fidelity must therefore be treated as coupled responses rather than independent goals.

Support-assisted annealing modifies the mechanical boundary condition during heating and cooling. Granular media can provide distributed resistance around complex surfaces, whereas a mechanical fixture applies contact at designed locations. Published approaches include sand, ceramic powder moulds, salt and gypsum moulds, liquid baths, elastomeric encapsulation and related methods [38]-[48]. These processes are not thermally equivalent. Support below melting is a constrained annealing treatment; salt treatment near melting can become a remoulding or densification process and must not be interpreted as the same mechanism. Support can reduce gross movement, but it can also change heating lag, friction, surface cleanliness, contact stress and cooling rate.

The central research question is: can support-assisted, sub-melting annealing improve or retain useful mechanical performance in FFF-printed commercial PLA while reducing the dimensional distortion produced by otherwise identical unconstrained annealing? Secondary questions concern temperature, duration, support method, interaction effects, statistical significance, practical effect size and industrial feasibility. The paper addresses these questions through a literature-supported two-stage design with one controlled print condition, paired oven-run blocks, three-axis metrology, tensile and flexural testing, and multi-response optimization. Its value lies in a comparable decision framework: constrained annealing itself has precedent, but a common low-temperature comparison of unconstrained treatment, multiple granular supports and a quantified fixture remains limited.

1.1 Research questions

1. How do annealing temperature and hold duration influence signed dimensional change, warpage, tensile strength, tensile modulus and strain at break in one controlled FFF PLA process?
2. Does physical support reduce dimensional distortion relative to unconstrained annealing at the same measured thermal exposure?
3. Which of sand, fine sodium chloride and a mechanical fixture offers the most useful combination of restraint, repeatability, cleanliness and safety at a common screening condition?
4. Do temperature x duration and constraint x thermal-condition interactions change the mechanical-dimensional trade-off?
5. Can a Pareto-efficient and independently confirmed processing window be identified without accepting a condition that violates a dimensional or safety limit?
6. How can a low-cost procedure be made reproducible in an undergraduate material-extrusion laboratory?

1.2 Contribution of the study

The contribution is a comparable decision framework, not a universal heat-treatment recipe. It fixes print architecture, separates a support screen from the factorial, distinguishes specimens from oven-run replicates, retains signed three-axis responses and requires confirmation. If no method passes the screen, that bounded negative result still defines the limits of low-cost restraint for the tested PLA.

2. Literature Review

2.1 FDM/FFF additive manufacturing

FFF and FDM commonly denote material extrusion of thermoplastic filament. A drive feeds filament into a hot end, and the nozzle deposits softened roads as perimeters, infill and successive layers. The part is an accumulation of roads, so its dimensional and mechanical performance depends on the temperature-time history of each road and interface [9]-[11].

Material-extrusion (FFF/FDM) process

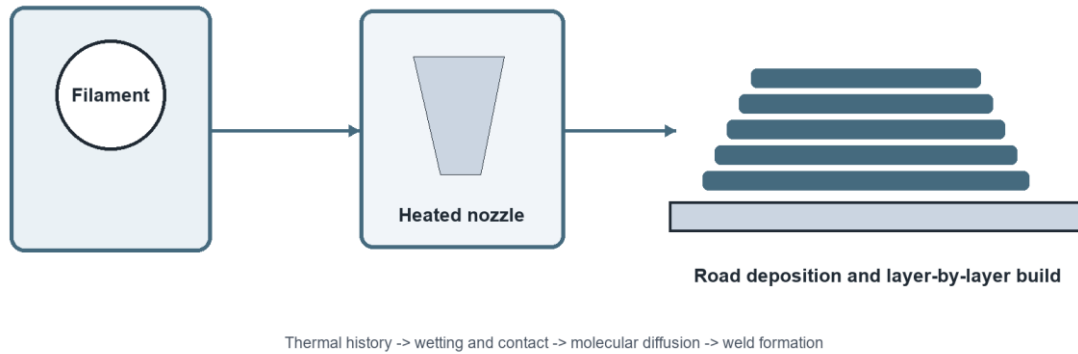


Figure 1. Original schematic of filament feeding, road deposition and layer-by-layer material extrusion. Not to scale.

2.2 PLA material behaviour

PLA grades vary in stereochemistry and molecular architecture. D-lactide content, chain length, branching, nucleation, plasticization and degradation history influence transitions, crystallization and mechanics [18]-[20]. Commercial filament can also contain undisclosed pigments or modifiers, while moisture promotes hydrolysis during heating. A valid study therefore retains one documented lot, controls storage and avoids importing thermal constants from another formulation.

Table 1. PLA thermal behaviour relevant to post-print annealing.

Thermal regime	Polymer process	Engineering implication	Sources
Below glass transition	Restricted segmental motion; printed orientation and residual stress are largely frozen.	Limited structural rearrangement over short holds; low distortion risk but modest annealing effect.	[18], [19]
Glass-transition region	Segmental mobility rises and stress relaxation becomes possible.	Shape movement can begin even though the polymer is far below melting; grade and history control the interval.	[18]-[20]
Cold-crystallization region	Previously amorphous chains may organize when time and mobility are sufficient.	Heat resistance or stiffness may increase, accompanied by anisotropic shrinkage or thickness change.	[20], [28]-[33]
Near melting and melting	Crystal order is lost and macroscopic flow or remoulding becomes possible.	Outside the intended factorial; near-melt salt methods must be discussed separately.	[18], [20], [40], [41]
Thermal degradation	Hydrolysis, oxidation and chain scission may alter mass, colour, odour and properties.	Dry handling, verified temperature, ventilation and conservative residence time are required.	[18]-[20]

Note: Transition temperatures are formulation- and history-dependent; the tested filament's supplier information and measured oven history take precedence over a generic value.

2.3 FDM process parameters

Layer height, nozzle temperature, orientation, raster, speed, cooling, infill and shell count alter voids and load transfer. Chacon et al. reported orientation-dependent layer-height and feed-rate effects [22]; Lanzotti et al. quantified layer, infill and perimeter contributions [23]; and Rajpurohit and Dave examined raster and layer variables [24]. Build orientation and infill effects can be strong [25], and strength, modulus and toughness need not share an optimum [26], [27]. Printing parameters must therefore be controlled inputs.

2.4 Interlayer bonding and anisotropy

A deposited road must conform to the previous layer and develop molecular interpenetration. Bellehumeur et al. modelled neck growth [12], Sun et al. linked processing to bond quality [13], and Coogan and Kazmer separated wetting, contact and diffusion [14]. Thermal studies show how rapidly weld temperature falls [15], while PLA work connects cyclic temperature, coalescence and residual alignment to interface strength [16], [17]. Annealing may relax orientation, but cannot be assumed to remove every void or raster-induced anisotropy.

2.5 PLA thermal behaviour

Printed PLA contains amorphous material, orientation and some crystalline order. Reheating enables segmental motion, possible cold crystallization and crystal reorganization [20], [28]-[32]. Crystalline form also matters [29]; prior processing changes kinetics [30]; and time, temperature, formulation and nucleation interact [31], [32]. Annealing response is therefore conditional rather than monotonic.

PLA response during sub-melting annealing

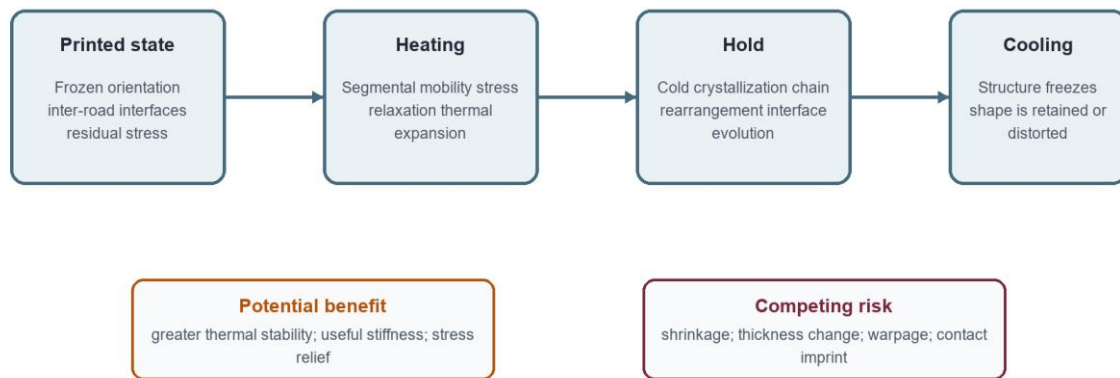


Figure 2. Competing molecular and geometric responses during sub-melting PLA annealing. Original technical schematic.

2.6 Annealing of PLA

FFF studies connect post-print heating with crystallinity, stress, interfaces and dimensions. Wach et al. combined structural and mechanical evidence [33]; Lee et al. studied stress relaxation and cooling [34]; Butt and Bhaskar reported material-dependent responses [35]; and von Windheim et al. separated porosity, crystallinity and adhesion [36]. Grade and orientation alter whether strength, stiffness or heat resistance improves [32], [37]. The thermal regime and material must therefore be identified explicitly.

2.7 Mechanical effects of annealing

Stress relaxation may reduce frozen stress, crystallization may raise stiffness and interfacial mobility may improve selected contacts. Conversely, crystallinity can restrict ductility, hydrolysis can lower molecular mass, and shape change alters the section used for stress [28]-[37]. Post-treatment width and thickness must therefore be measured, and strength, modulus and strain at break reported separately.

2.8 Annealing-induced dimensional distortion

Dimensional movement combines orientation recovery, residual-stress release, thermal expansion, crystallization-related volume change, gravity and friction. Raster architecture makes it anisotropic: one axis can contract while thickness grows, and gradients can bow the part. Stojkovic et al. evaluated tensile strength and dimensional accuracy together [42], while other PLA studies show condition-, orientation- and grade-dependence [37], [43], [44]. Length alone can hide width, thickness or flatness failure; three signed axes and warpage are required.

2.9 Constrained and support-assisted annealing

Support-assisted evidence includes oven, sand, water and glycerol treatments for tough PLA [38], ceramic-powder moulds for PLA and ABS [39], and salt or gypsum moulds in regimes that can approach melting [40], [41]. Further work compares oven and sand for filled systems [45], performs multi-parameter optimization [46], uses salt or elastomeric containment [47], and studies sand-embedded high-heat PLA [48].

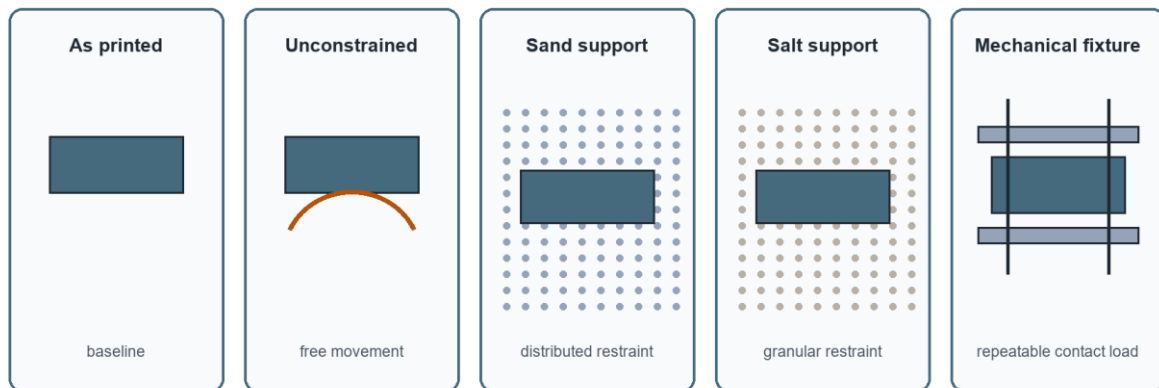
The support physics differ. Sand is conformable, but packing and moisture affect restraint and heating lag. Fine salt packs closely, yet is hygroscopic, corrosive when damp and distinct from near-melt remoulding. A fixture is reusable and measurable, but excessive load or mismatched expansion can imprint the surface or create stress. Screening is more defensible than assuming one method is best.

Table 2. Synthesis of previous annealing and support-assisted research.

Study	Material	Thermal/support condition	Principal relevance	Limitation for this study
Wach et al. (2018) [33]	FFF PLA	Annealing with structural and mechanical characterization	Links thermal treatment to crystallinity and mechanical response.	No common comparison of granular and fixture restraint.
Wijnbergen et al. (2021) [38]	Tough PLA	Oven, sand, water and glycerol	Treatment medium changes deformation and strength.	Grade and liquid-media physics limit transfer to neat commercial PLA.
Lluch-Cerezo et al. (2022) [39]	PLA and ABS	Ceramic powder mould	Powder support can limit deformation during thermal treatment.	No quantified mechanical fixture in the same design.
Szust and Adamski (2022) [40]	FFF polymers	Annealing and salt treatment near melting	Clarifies the distinction between annealing and remelting.	Near-melt effects cannot be assigned to low-temperature support.
Vorkapic et al. (2022) [41]	PLA and other thermoplastics	NaCl and gypsum moulds	Mould-assisted treatment changes tensile behaviour.	PLA regime approaches remoulding rather than only stress relief.
Stojkovic et al. (2023) [42]	FFF parts	Layer height x temperature x time	Studies tensile response and dimensional accuracy together.	Does not compare support boundary conditions.
Shbanah et al. (2023) [43]	Printed PLA	55-95 C with extended holds	Shows strong dependence on orientation and grade.	Physical support is not the principal factor.
Ghasemkhani et al. (2024) [44]	Printed PLA	Heat-treatment mechanisms	Discusses crystallization and interlayer-bonding competition.	Dimensional restraint is not systematically varied.
Butt et al. (2024) [45]	Fibre-filled systems	Oven versus sand	Sand response depends on composite formulation.	Not standard neat PLA and no fixture comparison.
Kahya et al. (2025) [46]	3D-printed PLA	70-110 C; 40-200 min; optimization	Demonstrates multi-parameter thermal optimization.	Objective structure differs and support is not the main factor.
Chiscop et al. (2025) [47]	PLA prints	Salt and RTV encapsulation	Containment can support dimensional objectives.	Application-specific encapsulation limits generalization.
de Assis et al. (2026) [48]	High-heat PLA	Sand-embedded conditioning	Recent direct mechanical-dimensional trade-off evidence.	High-heat formulation limits transfer to conventional PLA.

Note: The entries synthesize verified publications [33], [38]-[48]; they are not results obtained at DTU.

Annealing boundary conditions (schematic, not to scale)



Boundary conditions change motion, heat transfer, friction, contact stress and cooling history.

Figure 3. Comparison of as-printed, unconstrained, granular-supported and mechanically fixture-constrained boundary conditions. Original technical schematic.

2.10 Research gap

Supported annealing is not unstudied. The gap is comparability: published work varies PLA formulation, geometry, orientation, thermal regime, support physics and response definition, and some salt processes approach melting. The targeted literature did not identify one peer-reviewed study that screens sand, salt and a quantified fixture against unconstrained sub-melting treatment in one commercial PLA, then uses a replicated temperature x duration factorial with signed three-axis metrology, warpage, tensile response, block-aware statistics, multi-response optimization and confirmation.

3. Research Objectives

3.1 Primary objective

To evaluate whether physical support during sub-melting annealing can improve or retain useful mechanical performance of FFF-printed commercial PLA while reducing three-axis dimensional distortion and warpage relative to unconstrained annealing.

3.2 Secondary objectives

7. Explain the thermal, crystallization, stress-relaxation and interfacial mechanisms relevant to annealed FFF PLA.
8. Compare as-printed, unconstrained, sand-supported, salt-supported and gap-controlled fixture conditions at 95 C for 60 min.
9. Quantify temperature, hold-time and constraint effects at 80, 95 and 110 C for 30, 60 and 90 min.
10. Measure tensile strength, tensile modulus, strain at break, signed length/width/thickness change and out-of-plane warpage.
11. Use a limited three-group flexural comparison to check whether the tensile-based decision transfers to bending.
12. Separate treatment effects from print-plate, oven-run and measurement variation through randomization, blocking and repeatability checks.
13. Develop a transparent multi-response decision that maximizes useful strength and stiffness while minimizing dimensional error and warpage.
14. Assess low-cost implementation, contamination, cleaning, thermal lag, fixture reuse and laboratory safety for FFF applications.

4. Research Hypotheses

Table 3. Research hypotheses and corresponding null statements.

Pair	Research hypothesis	Null statement
H1 / H01	At least one support-assisted condition reduces dimensional error and warpage relative to thermally matched unconstrained annealing while retaining a predefined fraction of its tensile response.	No practically meaningful paired improvement is present within the declared margins.
H2 / H02	Temperature has a non-linear influence: increasing mobility and crystallization may initially benefit selected responses, while higher exposure increases distortion risk.	Temperature and curvature terms have no effect after accounting for other factors and blocks.
H3 / H03	Hold duration changes mechanical and dimensional responses until a kinetic saturation region is approached.	Duration has no effect within the tested range.
H4 / H04	Constraint changes the temperature and/or duration response because support modifies motion, friction and heat transfer.	Constraint x temperature and constraint x duration interactions are zero.
H5 / H05	One screened support provides a better mechanical-dimensional balance than unconstrained treatment under the predeclared selection rule.	No screened support improves that balance beyond uncertainty and practical limits.
H6 / H06	The selected processing condition is reproducible with specimens from new print and oven-run blocks.	Independent confirmation is inconsistent with the model interval or acceptance gates.

Note: Practical dimensional limits and tensile-retention criteria are to be agreed before the screen is evaluated; statistical significance alone does not define engineering usefulness.

5. Materials and Methods

5.1 Material

The material is commercial neat PLA filament, nominally 1.75 mm, from one colour and production lot. The actual trade name, lot, source, storage history and supplier thermal limits are recorded at procurement rather than guessed. Diameter is checked at distributed positions with a calibrated micrometer. The lot is stored sealed and dry; any approved drying procedure is applied uniformly. A new lot is not pooled unless it is deliberately modelled as a block.

5.2 FDM process

One calibrated material-extrusion system is used, with its model and firmware recorded from the laboratory machine. Controlled settings are a 0.40 mm brass nozzle, 210 C nozzle, 60 C plate, 0.20 mm layer, 100% rectilinear/lines infill, alternating +45/-45 degree raster, 45 mm s-1 nominal speed, flat XY orientation, constant validated cooling, three perimeters and five solid top/bottom layers. The slicer release, profile, CAD, mesh and G-code are archived.

Table 4. Controlled FFF printing parameters.

Variable	Specified condition	Control status	Engineering rationale
Material	Commercial neat PLA; one colour and lot	Fixed	Controls formulation and moisture history
Filament diameter	1.75 mm nominal; measured by micrometer	Recorded	Checks feed consistency
FFF system	One calibrated material-extrusion machine	Fixed	Avoids machine-to-machine thermal variation
Nozzle	0.40 mm brass	Fixed	Controls road scale and heat transfer
Nozzle temperature	210 C	Fixed	Controls melt state and inter-road welding
Build-plate temperature	60 C	Fixed	Controls first-layer adhesion and cooling
Layer height	0.20 mm	Fixed	Controls interface count and road geometry
Infill	100%; rectilinear/lines	Fixed	Limits changes in internal volume fraction
Raster sequence	+45/-45 degrees	Fixed	Creates a consistent balanced in-plane layup
Nominal print speed	45 mm s-1	Fixed	Controls deposition and cooling history
Build orientation	Flat XY	Fixed	Controls anisotropy and specimen placement
Cooling	Same validated fan profile	Fixed and archived	Controls solidification and initial structure
Shells	3 perimeter walls; 5 solid top and bottom layers	Fixed	Controls shell fraction and edge load path
Software	One recorded slicer release and archived profile	Fixed	Preserves digital process definition

Note: The printer model, firmware, filament trade name and slicer release are recorded from the actual laboratory resources; they are not guessed in this paper.

5.3 Specimen geometry

Tensile coupons follow ASTM D638-22 Type I where machine and build capacity permit: 165 mm minimum overall length, 13 mm narrow width, 57 mm narrow length, 50 mm gauge length, 115 mm nominal grip separation and 3.2 mm nominal thickness [4]. If Type IV is necessary, it is adopted for every tensile group before printing. ASTM D790-25 flexural bars use a nominal 127 x 12.7 x 3.2 mm planning geometry [5]. Actual measured sections govern stress calculations. All coupons are flat XY and reported using ISO 17295:2023 [2].

5.4 Printing and specimen allocation

Identifiers are assigned before printing, and conditions are distributed across plates and randomized positions so treatment is not identical to print batch. Records retain plate ID, position, time, operator, ambient conditions and defects. Removal and stabilization are consistent. A failed print is documented and replaced with a new identifier, never silently excluded.

5.5 Specimen conditioning

Specimens are conditioned under ASTM D618-21 Procedure A for at least 40 h at 23 +/- 2 C and 50 +/- 10% relative humidity where applicable [6]. The same atmosphere is used after treatment and during testing. Time, atmosphere and deviations are recorded because PLA dimensions and properties are moisture- and temperature-sensitive.

5.6 Baseline measurements

Before heating, overall length, gauge width/thickness, mass and maximum out-of-plane warpage are measured at fixed locations. ASTM D5947-24 guides solid-plastic dimensions [7]. Instrument resolution must be small relative to the expected change. Three repeated placements of representative coupons establish short-term metrology repeatability and standardize measurement force and orientation.

Dimensional and warpage measurement

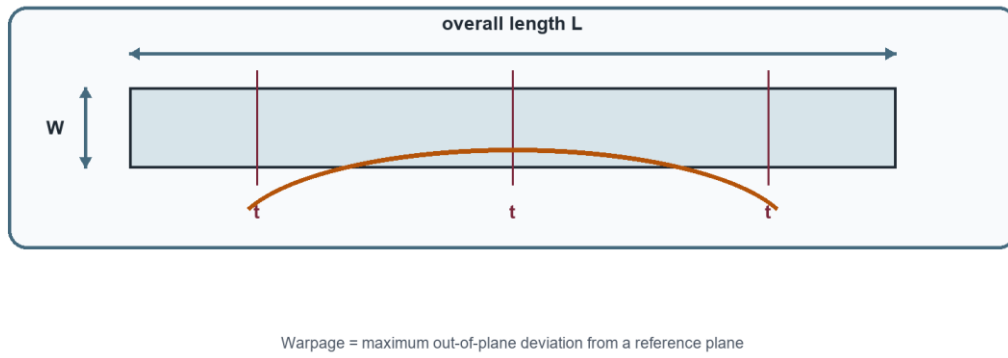


Figure 4. Measurement locations for length, width, thickness and out-of-plane warpage. Original technical schematic.

$$\Delta L (\%) = [(L_{\text{after}} - L_{\text{before}}) / L_{\text{before}}] \times 100 \quad (1)$$

Analogous signed changes are calculated for width and thickness. Positive values denote expansion and negative values denote contraction. Signed values are retained even when a scalar dimensional-error response is used.

$$E_{\text{RMS}} (\%) = \sqrt{[(\Delta L)^2 + (\Delta W)^2 + (\Delta t)^2] / 3} \quad (2)$$

If drawing tolerances are available, a tolerance-normalized error is preferred because the allowable deviation may differ by axis.

$$E_{\text{tol}} = \sqrt{\{[(L_{\text{after}} - L_{\text{before}}) / T_L]^2 + [(W_{\text{after}} - W_{\text{before}}) / T_W]^2 + [(t_{\text{after}} - t_{\text{before}}) / T_t]^2\} / \sqrt{3}} \quad (3)$$

5.7 Annealing setup

A dedicated ventilated oven receives a spatial-uniformity check. A calibrated probe/logger is placed beside the coupon or in a sacrificial support assembly. After the empty oven stabilizes, the tray is loaded and the hold starts only when the specimen environment returns within +/- 2 C of target. The log retains heat-up delay, temperature range, excursions and door openings. Treatments are 80, 95 or 110 C for 30, 60 or 90 min [37], [42], [43], [46]. The 110 C level proceeds only after a sacrificial coupon shows no sticking, collapse, smoke or abnormal odour.

After the hold, the hot tray is moved to a heat-resistant surface. The specimen remains constrained until the adjacent probe is below 50 C, then is released and cleaned. Cooling location, release time and movement are recorded. Early release can permit distortion, while slow cooling can alter crystallization; measured specimen temperature, not display setpoint alone, defines exposure.

5.8 Sand support

Low-dust graded sand is dried to constant mass at 105 +/- 5 C and cooled sealed. A lidded heat-stable tray receives a 20 mm base; the oriented specimen is covered by at least 20 mm and given one fixed gentle vibration cycle without tamping the gauge. Grade, mass, fill height, vibration and clearance are recorded because packing and thermal lag are part of the process.

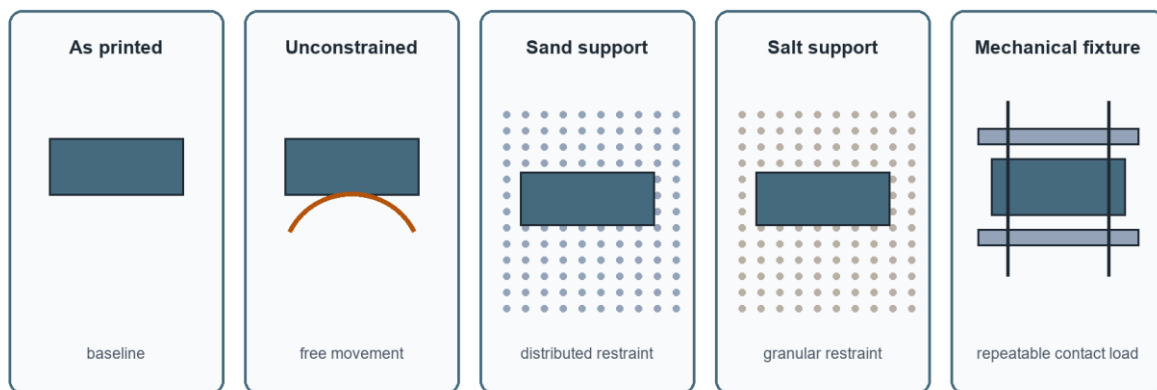
5.9 Salt support

Fine NaCl is screened only as dry support below melting. It is dried to constant mass at 105 +/- 5 C, cooled sealed, and packed in a dedicated corrosion-resistant tray with the same 20 mm base/cover depths. Grade, mass, fill and moisture history are recorded. After cooling, salt is removed without prolonged water exposure and cleaning is documented. This differs explicitly from near-melting remoulding [40], [41]; damp salt and vulnerable metals are avoided.

5.10 Mechanical fixture

Two flat, thermally stable plates use PTFE release films and four symmetric fasteners. Precision stops set the hot gap to measured specimen thickness plus 0.05 mm, limiting bow without intentional gauge compression. Fasteners seat against the stops diagonally; plate material, flatness, stop thickness and seating-force calibration are recorded. Each run is inspected for expansion, bow, imprinting and edge contact. Gap control is more reproducible than arbitrary torque.

Annealing boundary conditions (schematic, not to scale)



Boundary conditions change motion, heat transfer, friction, contact stress and cooling history.

Figure 5. Boundary-condition details for the support-method screen. The same diagram is repeated here to connect the literature concepts with the laboratory procedure.

5.11 Experimental matrix

Stage 1 screens AP, U, sand, salt and fixture conditions at n=5; annealed groups use 95 C/60 min. Selection applies safety and feasibility gates before dimensional error, warpage, tensile retention, variability and cleaning. Stage 2 is a 3 x 3 x 2 factorial of temperature, duration and U versus selected constraint, n=5 per cell. AP and the applicable 95 C/60 min screen groups are reused with full traceability.

Table 5. Experimental factors, levels and control structure.

Factor	Levels	Role	Control and interpretation
Stage 1 method	AP; U; sand; salt; fixture	Five levels	Screen at 95 C for 60 min; AP receives no heating
Temperature	80; 95; 110 C	Three levels	Verified specimen-environment temperature
Duration	30; 60; 90 min	Three levels	Hold starts inside +/- 2 C setpoint band
Definitive constraint	U; selected support	Two levels	Selected after the screen using declared gates
Replication	n = 5 per condition	Independent specimens	Paired across five separate oven-run blocks

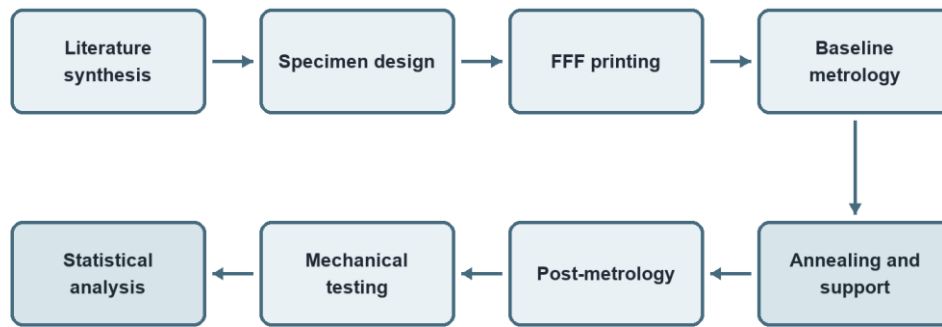
Factor	Levels	Role	Control and interpretation
Print nuisance factors	Plate, position, print time	Recorded/blocked	Prevent treatment from being identical to print batch
Thermal nuisance factors	Oven run and position	Recorded/random	Separate run variation from specimen residual error

Table 6. Specimen plan and accounting.

Phase	Structure	Physical specimens	Interpretation
Constraint screen	5 methods x 5	25 tensile/dimensional	Includes AP, U and three support candidates at the centre condition
Definitive additional	16 non-centre cells x 5	80 tensile/dimensional	Completes the 18 annealed cells after reusing centre U and selected-support groups
Definitive dataset	18 annealed cells x 5 + AP x 5	95 tensile/dimensional	Primary factorial dataset; 15 records originate in the screen
Tensile confirmation	2 groups x 5	10 tensile/dimensional	New selected optimum and thermally matched U groups
Flexural confirmation	3 groups x 5	15 flexural	AP, matched U and selected optimum
Total physical specimens	25 + 80 + 10 + 15	130	115 tensile/dimensional plus 15 flexural

Note: Five coupons placed together in one oven cycle would be subsamples, not five thermal replicates. The definitive paired design therefore uses five independent oven-run blocks for each temperature-duration cell.

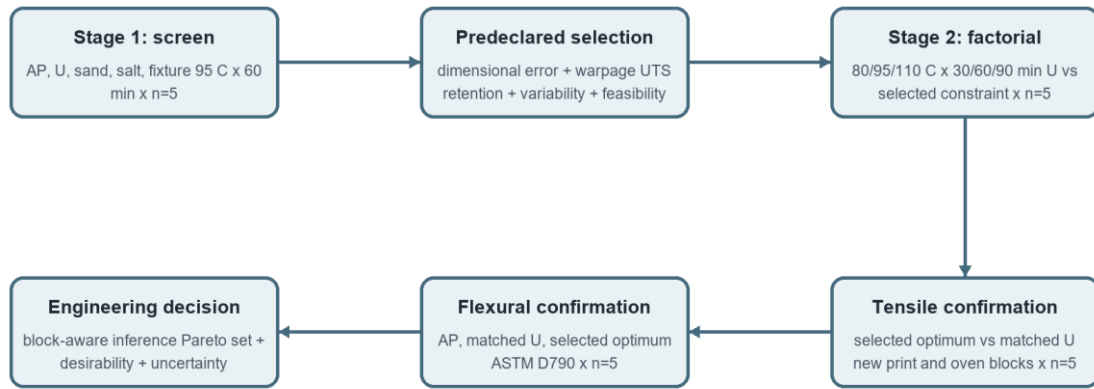
Research methodology and traceability



Trace every observation to specimen ID, print plate, oven run, position and instrument file.

Figure 6. Research methodology and data-traceability workflow. Original technical schematic.

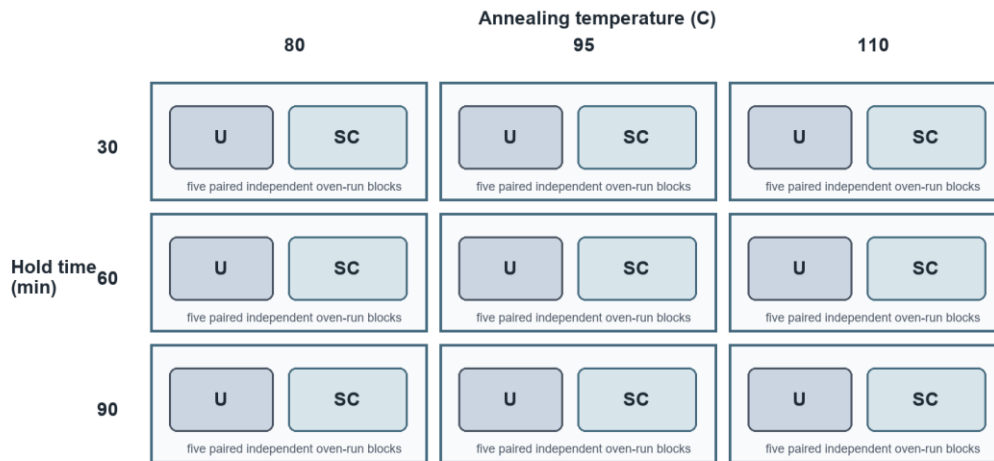
Two-stage experimental design



Total physical specimens: 130 (115 tensile/dimensional and 15 flexural).

Figure 7. Two-stage support screening, factorial study and confirmation logic. Original technical schematic.

Definitive temperature x time x constraint matrix



U = unconstrained; SC = selected constraint. Common as-printed baseline: n=5.

Figure 8. Definitive 3 x 3 x 2 factorial and paired oven-run blocking structure. Original technical schematic.

5.12 Tensile testing

ASTM D638-22 tensile tests use a calibrated UTM, suitable grips, load cell and extensometer [4]. Type I speed is 5 mm min⁻¹ unless the complete method requires otherwise. Post-treatment gauge width/thickness define the initial area. Report ultimate strength, extensometer modulus, strain at break, break location and failure mode. Crosshead travel does not replace extensometer strain for modulus; invalid failures follow a written rule and remain traceable.

5.13 Flexural testing

ASTM D790-25 flexure uses three-point loading [5]. Actual mean depth determines the nominal 16:1 support span and test rate. Report the deflection method because crosshead and deflectometer values differ. AP, thermally matched U and the selected condition are compared at n=5; this confirms transfer to bending rather than repeating the full optimization.

5.14 Dimensional measurement

After cooling and reconditioning, all dimensions, mass and warpage are remeasured at identical locations. Calibration and resolution are recorded. Duplicate readings are taken independently and a third resolves disagreement beyond repeatability. Report signed axis changes with E_RMS or E_tol, and report warpage with its reference plane and placement repeatability.

5.15 Optional thermal and microscopic characterization

Optional DSC of representative AP, U and selected-support samples follows ASTM D3418-21 with mass, pan, atmosphere, rate and cycle reported [8]. First heating captures the print/anneal history; a later heating is not the original state. Report transitions and integration limits. For neat PLA, crystallinity may use the commonly adopted 93.0 J g⁻¹ fully crystalline reference [21], with baseline and crystal-form uncertainty stated.

$$X_c (\%) = [(\Delta H_m - \Delta H_{cc}) / (w_{PLA} \times \Delta H_m^0)] \times 100 \quad (4)$$

Here ΔH_m is melting enthalpy, ΔH_{cc} is cold-crystallization enthalpy expressed as a positive magnitude, w_{PLA} is PLA mass fraction and ΔH_m^0 is the declared fully crystalline reference. No crystallinity value is calculated without genuine DSC data. Optical microscopy or SEM may compare voids, fracture surfaces and inter-road interfaces only when preparation, magnification, scale bars and sampling locations are documented.

5.16 Standards and measurement methods

Table 7. Standards used for specimen preparation, testing and reporting.

Purpose	Reference	Application in this investigation
AM terminology	ISO/ASTM 52900:2021 [1]	Use standard material-extrusion and FFF terminology
Part orientation	ISO 17295:2023 [2]	Report build coordinates and specimen orientation
Polymer AM reporting	ASTM F3498-23 [3]	Describe material handling, anisotropy, porosity and specimen location
Tensile properties	ASTM D638-22 [4]	Control specimen type, dimensions, speed, extensometry and validity
Flexural properties	ASTM D790-25 [5]	Use actual depth for span/rate and report deflection method
Conditioning	ASTM D618-21 [6]	Apply and report conditioning atmosphere and time
Dimensions	ASTM D5947-24 [7]	Standardize measurement of solid-plastic specimen dimensions
Optional DSC	ASTM D3418-21 [8]	Report transition and enthalpy method, cycle and integration

Note: The complete copyrighted standards control the test; this table is a planning summary and not a substitute for laboratory access to the current documents.

5.17 Response variables

Table 8. Response variables and evidence hierarchy.

Priority	Response	Unit	Measurement and interpretation
Primary mechanical	Ultimate tensile strength	MPa	Maximum engineering stress from measured post-treatment section
Primary dimensional	Signed Delta L, Delta W, Delta t	%	Before/after matched metrology
Primary dimensional	E_RMS or E_tol	% or dimensionless	Combined error, reported with all signed axes
Primary geometric	Warpage	mm	Maximum out-of-plane deviation from reference plane
Secondary mechanical	Tensile modulus	GPa	Extensometer-based initial stiffness
Secondary mechanical	Strain at break	%	Extensometer or validated strain method
Confirmation	Flexural strength and modulus	MPa; GPa	Three-group ASTM D790 comparison
Process quality	Mass change and temperature history	%; C; min	Balance and independent thermal logger
Optional thermal	Tg, Tcc, Tm, enthalpies, Xc	C; J g ⁻¹ ; %	DSC with declared baselines and cycles

5.18 Statistical analysis

Report individual data, mean, standard deviation, coefficient of variation and two-sided 95% confidence intervals. Stage 1 compares each support with U using multiplicity control and effect sizes. The definitive

model includes temperature, duration, constraint, required interactions, curvature and oven run as a block/random effect. Paired U-supported coupons improve the constraint contrast; print plate and position remain nuisance terms. Randomization, replication and blocking follow DOE principles [49].

Predeclared tests use $\alpha = 0.05$ with residual, normal-quantile and influence diagnostics. Transformation, heteroscedastic models or robust sensitivity checks require justification. Report exact p-values with effects and intervals. Statistical significance is not manufacturing importance: a dimension outside tolerance or unacceptable warpage fails regardless of strength. Five independent specimens per cell mainly resolve large effects; add blocks if key contrasts remain imprecise.

5.19 Safety and feasibility

Essential equipment is a calibrated FFF printer, PLA lot, ventilated oven, independent logger, calibrated caliper/micrometer, surface plate and warpage indicator, balance, UTM with extensometer, heat-stable trays and support/fixture hardware. A flexural fixture is strongly recommended. DSC, TGA, microscopy, SEM and profilometry are optional.

Use thermal gloves, tongs, eye protection and heat-resistant surfaces. Smoke, sticking, strong odour or abnormal colour stops oven work. Control sand dust; keep salt dry and away from vulnerable metals; guard fixture pinch points. Mechanical testing requires correct grips/load cell, eye protection and a fragment exclusion zone. Local DTU risk assessment and supervisor instructions take precedence.

6. Results and Discussion - Literature-Based Engineering Analysis

This section analyses published experiments to establish mechanisms, feasible conditions and competing trends. Cross-study numerical comparison is limited by different PLA grades, geometries, orientations and response definitions. Any numerical value is explicitly source-attributed and is not a measurement of the planned DTU coupons.

6.1 Effect of annealing temperature

Temperature controls the rate of relaxation and crystallization. At 80 C, many commercial grades are mobile without approaching melting; gross distortion may be lower, but short holds may not strongly crystallize a slow grade. Even modest treatment can alter dimensions while overall form is retained [34], [37], [42], [43].

At 95 C, mobility is greater and the level is represented in extended-time and optimization work [43], [46]; it is a discriminating screen condition, not a declared optimum. At 110 C, crystallization may accelerate, but so can stress release, contact marking and gradients. Kahya et al. tested 70-110 C and 40-200 min, reporting 90 C/120 min as favourable for their material [46]. This supports a central region while showing why another grade cannot inherit that optimum.

A non-linear response is expected: selected properties may improve initially, while dimensional error accelerates with mobility. Three levels can reveal a middle compromise that two levels miss. An unsafe 110 C pilot requires a revised safe window, not continued testing for statistical completeness.

6.2 Effect of annealing duration

Duration is measured after the specimen environment reaches setpoint, because supports create thermal lag. Thirty minutes may allow early relaxation with limited movement; 60 min is a practical screen centre; and 90 min tests continued change or saturation.

Longer is not always better: crystallization can plateau while distortion or degradation risk accumulates. Extended 55-95 C treatments show orientation and grade dependence [43], and nucleation alters response [32]. Duration must therefore interact with temperature and support-dependent heating lag.

6.3 Effect on tensile strength

Tensile findings are mixed because interface rearrangement and stress relief can improve load transfer, while voids, degradation or restricted ductility can reduce performance [33]-[37], [44]. Annealing also changes the stress denominator; using nominal rather than measured post-treatment dimensions can create a false strength increase.

Wijnbergen et al. compared four media and found method- and orientation-dependent deformation and strength [38]. Sand produced the least ring deformation: 2.04% inner-diameter shrinkage and 1.99% height increase for horizontal rings. These source-specific, opposing signs show why one-dimensional shrinkage is inadequate. Other mould and sand studies also find containment-dependent mechanics, with limited transfer across grades and regimes [39], [41], [45].

6.4 Effect on tensile modulus and ductility

Modulus can rise with crystalline order while strength remains controlled by voids and fracture paths; strain at break may fall as local strain redistribution becomes harder [28], [29], [32]. This matters for jigs: stiffness can improve positioning while brittleness worsens clamping or impact behaviour.

Extensometer strain is required for modulus, and ductility needs a common validity rule. A specimen that breaks outside the gauge after fixture imprinting is not evidence of improved bulk PLA. Failure locations must accompany summary statistics.

6.5 Dimensional shrinkage and warpage

Dimensional change is structural recovery. Nozzle flow, toolpath turning, flattening and non-uniform cooling leave direction-dependent orientation and stress. Reheating can contract a preferred road direction while width or thickness grows; crystallization adds volume change, and gradients, gravity and friction convert recovery into curvature.

Printing and annealing variables jointly affect tensile and dimensional response [42], while orientation, architecture and grade can reverse trends [37], [43], [44]. A specimen may retain length while bowing or changing section. Delta L, Delta W, Delta t, combined error and direct warpage are therefore all retained.

6.6 Effect of sand support

Sand provides distributed, conformal restraint without a custom mould. It minimized deformation for the tough-PLA ring study [38]; powder restraint is also plausible [39]; and oven-versus-sand response depends on composite formulation [45]. These findings justify screening sand but do not prove superiority for neat PLA coupons.

Granular support also varies with particle size, moisture, mass, cover depth and vibration. Denser packing may restrain more but delay heating or create pressure gradients; dust and cleaning add burden. The medium, packing and measured specimen temperature must therefore be recorded.

6.7 Effect of salt support

Dry fine salt fills small gaps and is inexpensive. Published methods show that salt containment can preserve shape or support more severe treatment [40], [41], [47], but some approach melting or autoclave conditions. Their mechanics do not transfer directly; this screen asks only whether dry salt restrains sub-melting movement.

Salt absorbs moisture, contaminates surfaces, corrodes when damp and may require washing that changes conditioning. It should advance only if repeatable benefit justifies dedicated drying, trays and cleaning; a small geometric gain with unreliable moisture history is not industrially attractive.

6.8 Effect of mechanical constraint

A fixture provides a defined reusable boundary: plates suppress bow, stops define the gap, and release films permit in-plane slip. It can be calibrated and rapidly reused. Mould and encapsulation studies exist [39], [41], [47], but a low-temperature comparison adding a quantified fixture to sand, salt and U remains limited.

Over-restraint can store stress, buckle edges, emboss surfaces or transfer fixture expansion into PLA. Poor plate flatness can exchange global warp for local thickness error. Gap control avoids intentional compression, and imprint/contact damage fails the feasibility gate. Flatness alone cannot select the fixture if tensile scatter or surface integrity worsens.

Table 9. Literature-based comparison of annealing responses in FFF PLA.

Reference	Material	Thermal condition	Support	Mechanical response	Dimensional response	Engineering conclusion
Wach et al. [33]	FFF PLA	Study-specific thermal annealing	No external support reported	Mechanical and structural response changed with treatment.	Dimensional control was not the main comparison.	Thermal history must be linked to both structure and mechanical response.
Wijnbergen et al. [38]	Tough PLA	Oven, sand, water, glycerol	Four media	Strength response depended on method and print direction.	Sand gave the least ring deformation; horizontal rings showed 2.04% inner-diameter shrinkage and 1.99% height increase.	Support medium and orientation must be treated as process variables.
Lluch-Cerezo et al. [39]	PLA and ABS	Multiple thermal conditions	Ceramic powder mould	Mechanical response was material- and treatment-dependent.	Powder moulding limited deformation under selected conditions.	Powder support is plausible but not universal across polymers.
Vorkapic et al. [41]	PLA and other thermoplastics	Includes near-melting treatments	NaCl and gypsum moulds	Mould treatment altered tensile performance.	Containment enabled shape control in the studied regime.	Near-melt remoulding must be separated from sub-melting annealing.
Stojkovic et al. [42]	FFF specimens	Temperature-time factorial with layer height	Unconstrained	Tensile strength varied with the combined process factors.	Dimensional accuracy also changed and required joint evaluation.	Optimizing strength alone can select an inaccurate condition.
Shbanah et al. [43]	Printed PLA	55-95 C with extended holds	Unconstrained	Mechanical response depended on orientation and condition.	No common physical-restraint comparison.	The print architecture limits transfer of a thermal result.
Kahya et al. [46]	3D-printed PLA	70-110 C; 40-200 min	Unconstrained optimization	Reported 90 C/120 min as the favourable combined mechanical setting in that design.	Dimensional restraint was not the principal objective.	Multi-response settings are local to material, factors and weights.
de Assis et al. [48]	High-heat PLA	Sand-embedded thermal conditioning	Sand	Reported a direct mechanical response to supported conditioning.	Dimensional response was evaluated with the high-heat formulation.	Recent evidence supports the trade-off question but not transfer to neat PLA.

Note: Only the Wijnbergen et al. dimensional values are quoted numerically because they were verified from the source. All other entries are qualitative syntheses and must not be read as measurements of the planned DTU specimens.

Table 10. Literature-supported trends to be tested experimentally.

Parameter change	Expected trend	Engineering reason	Required caution
Temperature rises within a safe sub-melting window	Mobility and crystallization rate generally increase initially.	Segmental motion and chain organization become faster.	Non-linear response; high levels may distort
Excessive thermal exposure	Dimensional error, warpage or contact imprint may increase.	Stress release, softening, gradients and gravity permit movement.	Pilot stop rule and direct metrology
Duration increases from 30 to 90 min	Structural change may grow and then approach saturation.	Crystallization and relaxation have finite kinetics.	Duration x temperature interaction
Sand support	Gross deformation may decrease relative to U.	Distributed granular reaction resists motion.	Packing variability, dust and thermal lag
Salt support	Granular restraint may reduce movement.	Fine particles surround the surface.	Moisture, corrosion and cleaning
Gap-controlled fixture	Flatness control may be strongest for planar coupons.	Direct geometric restraint limits out-of-plane motion.	Imprint, stored stress and thermal expansion
Constraint becomes excessive	Local thickness error or brittle failure may increase.	Contact stress and locked recovery create new stress concentrations.	Inspect surface and retain failure mode

Note: These are hypotheses derived from published mechanisms and studies; they are not observed DTU outcomes.

7. Multi-Response Optimization Framework

Tensile strength alone is an incomplete decision: a stronger part may exceed tolerance, warp or become brittle, while minimum distortion may come from an ineffective thermal cycle. The objectives therefore maximize useful strength/stiffness and minimize E_{RMS} or E_{tol} , warpage and shrinkage. Strain at break has a lower-bound desirability; safety, sticking, collapse and surface damage are feasibility gates.

Each response y_i is converted to desirability d_i from 0 (unacceptable) to 1 (target). Maximize functions rise from a minimum acceptable value; minimize functions fall toward a maximum tolerable value. Limits, weights and curve shapes are set before outcome review under the Derringer-Suich formulation [50].

$$D = [\text{product}(d_i \wedge w_i)] \wedge [1 / \text{sum}(w_i)] \quad (5)$$

Here d_i is an individual desirability and w_i its predeclared weight. The geometric mean makes D zero when any response is unacceptable, so strength cannot hide dimensional failure. Report D with each d_i and raw units; vary reasonable weights and limits to test selection stability.

Pareto analysis complements D. A condition is dominated when another is no worse on every response and better on one. The efficient set exposes the strength-dimensional trade-off; selection occurs only after safety and tolerance gates. Literature suggests a central thermal region and controlled fixture or sand restraint may be favourable, but this is a testable expectation, not an optimum claim.

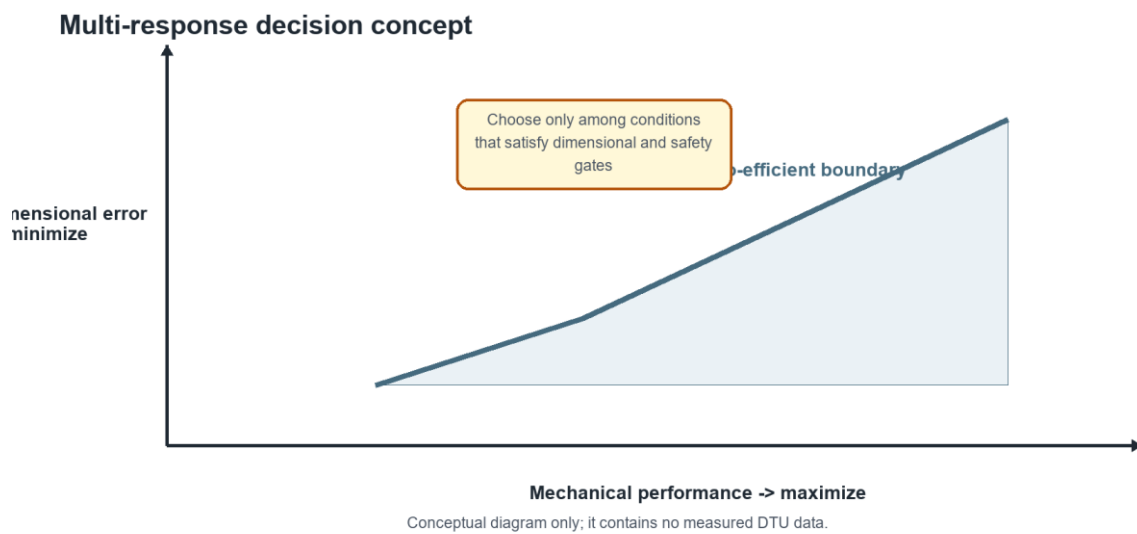


Figure 9. Conceptual Pareto and desirability decision framework. The diagram contains no experimental points or simulated measurements.

Confirmation uses new print and oven blocks: the selected support and thermally matched U at $n=5$ each. Responses must fall within prediction intervals and pass engineering gates; flexural groups check transfer to bending. Failed confirmation triggers review of model and boundary control rather than an optimum label.

8. Industrial Significance

A reliable support-assisted cycle could extend the usefulness of PLA in functional prototypes, assembly aids, jigs, fixtures, drill or inspection guides, low-volume tooling, replacement parts and components exposed to moderate heat. These applications value final geometry and repeatability as strongly as coupon strength. A jig that becomes stiffer but moves a locating surface outside tolerance has failed its purpose. The proposed response hierarchy reflects this reality by treating mechanical performance and dimensional accuracy simultaneously.

8.1 Advantages of constrained annealing

The principal attraction is a potentially low-cost post-process that uses existing FFF equipment and a controlled oven. Distributed or fixture restraint may reduce gross movement, improve part-to-part consistency, limit redesign iterations and make thermal conditioning more practical for tolerance-sensitive parts. Sand and salt can conform to irregular forms without machining a mould. A fixture can be reused and integrated with inspection for repeated geometries. The method also creates a disciplined qualification route: define the drawing need, screen boundary conditions, measure the thermal cycle, model a safe window and independently confirm the selected condition.

Table 11. Industrial comparison of annealing boundary conditions.

Method	Advantages	Limitations	Most plausible use	Relative implementation burden
Unconstrained	Lowest tooling cost; simplest cleaning	Free shrinkage and warpage; contact-face dependence	Exploratory coupons and geometries with generous tolerance	Low
Sand	Conformable; inexpensive; distributed restraint	Packing variability; dust; cleaning; thermal lag	Irregular low-volume parts when cleaning is accessible	Low to moderate
Salt	Fine packing; inexpensive; potentially dissolvable	Moisture; corrosion; contamination; washing control	Selected enclosed forms with dedicated corrosion-resistant handling	Moderate

Method	Advantages	Limitations	Most plausible use	Relative implementation burden
Gap-controlled fixture	Repeatable geometry; reusable; fast changeover for families	Design effort; thermal expansion; contact imprint; limited geometric flexibility	Repeated jigs, guides and planar tolerance features	Moderate initially; low per cycle

Note: The comparison is an engineering screening aid. Final selection remains local to the tested PLA, geometry, tolerance and cleaning requirements.

Transfer beyond coupons requires restraint geometry and thermal lag to be reconsidered. Thick ribs, cavities and topology-optimized forms do not heat or recover like a 3.2 mm bar. A production procedure must specify the part family, print architecture, support preparation, measured thermal window, release temperature, cleaning, inspection and acceptance rule. At minimum, a new filament grade, new wall thickness or new fixture material requires confirmation rather than direct extrapolation.

9. Limitations

The framework deliberately focuses on one commercial neat PLA lot, one printer, one slicer profile, flat coupon geometries and one raster architecture. PLA grades differ widely, so conclusions cannot automatically extend to tough, nucleated, high-heat, recycled, pigmented or filled products. Coupon response does not fully represent thick walls, sharp thickness transitions, enclosed cavities, assemblies or topology-optimized forms. The 80-110 C range is scientifically plausible but still requires a sacrificial safety pilot for the actual filament and oven.

Five independent specimens per cell are feasible for an undergraduate project but provide limited power for small interactions. Additional oven-run blocks may be necessary if residual variation is high. Oven uniformity, time to specimen setpoint, granular packing and fixture flatness can introduce variation even with a written method. The support screen selects one method for the factorial, so the definitive design does not estimate the complete temperature dependence of every support. Flexural testing is confirmatory and does not replace fatigue, creep, impact or heat-deflection testing.

Without DSC or microscopy, crystallization and interface explanations remain literature-supported mechanisms rather than direct measurements in the tested coupons. Even with DSC, a small cut sample may not represent through-thickness or between-road variation. Short-term post-treatment dimensions do not establish long-term aging, moisture response or creep under service load. Finally, no local effect estimate, p-value or optimum can be stated until specimens have been produced and tested under the documented procedure.

10. Future Scope

The most useful next step after the coupon study is geometry-dependent validation. Plates, rings, ribs, holes, bosses and mixed-thickness components should be annealed under the selected method and inspected with coordinate metrology or structured-light scanning. A dimensional compensation model could then relate stable signed shrinkage to CAD oversizing: print the predicted pre-compensated geometry, apply the controlled annealing cycle and confirm the final drawing dimensions. This extension should be evaluated with prediction intervals and independent parts before it is used as a manufacturing rule.

Material extensions include PETG, ABS, ASA, nylon, fibre-filled PLA, high-heat PLA and recycled formulations. Process extensions include constraint pressure or gap as a designed factor, heat-transfer modelling inside granular media, topology-optimized fixtures, controlled heating/cooling ramps, digital image correlation, computed tomography, heat-deflection testing, fatigue, creep and environmental aging. DSC, TGA, SEM and fracture microscopy could connect macroscopic response with crystallinity, degradation, voids and road interfaces. Response-surface or Bayesian sequential design may refine a safe window after the present factorial establishes curvature and interaction structure; machine-learning prediction is appropriate only when a sufficiently large, traceable, geometry-diverse dataset exists.

11. Conclusion

Annealing of FFF-printed PLA is a coupled thermal, structural and geometric process. Heating above the glass-transition region increases chain mobility and can promote stress relaxation, crystallization and changes at inter-road interfaces. Those mechanisms may improve heat resistance, stiffness or selected mechanical properties, but they also release the frozen orientation and residual stresses created during deposition. The

resulting contraction, transverse growth and warpage can make a nominally stronger component unsuitable for a tolerance-sensitive application. The literature therefore supports evaluating mechanical performance and dimensional stability together rather than ranking conditions by tensile strength alone.

Physical support offers a plausible means of controlling that trade-off, but sand, salt and mechanical fixtures impose different boundary conditions. Sand is conformable and inexpensive, yet its packing and thermal lag must be controlled. Salt can provide fine granular restraint but introduces moisture, corrosion and cleaning concerns and must be distinguished from near-melting salt remoulding. A gap-controlled fixture provides the most explicit and reusable geometric restraint for planar specimens, while risking contact imprint, stored stress and thermal-expansion mismatch. Existing studies demonstrate that supported treatments are feasible; they do not provide one universal best method because formulation, geometry, orientation and thermal cycle differ.

The research design resolves this uncertainty in two stages. A 25-specimen centre-condition screen compares AP, U, sand, salt and fixture boundary conditions at 95 C for 60 min. The selected feasible support then enters a replicated 80/95/110 C x 30/60/90 min factorial against U. Reuse of the appropriate screen coupons, followed by new tensile and flexural confirmation groups, gives 130 physical specimens. Standards-based specimen conditioning, measured three-axis dimensions, direct warpage, extensometer tensile testing, block-aware models, confidence intervals and confirmation protect the comparison from nominal-dimension errors and oven-cycle pseudoreplication.

The multi-response framework is central to the engineering decision. Pareto analysis retains the visible trade-off, while predeclared Derringer-Suich desirability combines acceptable mechanical and dimensional responses without allowing a severe failure to be hidden by an average score. No local optimum or property improvement is asserted before testing. Instead, the completed paper provides a technically defensible method and a literature-grounded expectation that an intermediate thermal exposure, combined with controlled physical restraint, may offer a more useful balance than either mild untreated behaviour or severe unconstrained heating. Execution of the documented experiments is required before the condition can be recommended for DTU laboratory or industrial use.

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Appendix A. Complete Experimental Matrix

The definitive dataset contains a common AP baseline and 18 annealed temperature-duration-constraint cells. Each row contains five independently printed coupons. For each temperature-duration cell, the five U coupons and five selected-constraint coupons are processed as pairs across five separate oven-run blocks. The 95 C/60 min U and selected-constraint groups originate in the Stage 1 screen and retain their original immutable specimen IDs.

Table A1. Definitive tensile and dimensional matrix.

Cell code	Temperature (C)	Duration (min)	Constraint	n	Specimen source
AP	None	None	None	5	Baseline; reused from Stage 1
T80-30-U	80	30	U	5	Definitive additional group
T80-30-SC	80	30	SC	5	Definitive additional group
T80-60-U	80	60	U	5	Definitive additional group
T80-60-SC	80	60	SC	5	Definitive additional group
T80-90-U	80	90	U	5	Definitive additional group
T80-90-SC	80	90	SC	5	Definitive additional group
T95-30-U	95	30	U	5	Definitive additional group
T95-30-SC	95	30	SC	5	Definitive additional group
T95-60-U	95	60	U	5	Stage 1 centre group
T95-60-SC	95	60	SC	5	Stage 1 centre group
T95-90-U	95	90	U	5	Definitive additional group
T95-90-SC	95	90	SC	5	Definitive additional group
T110-30-U	110	30	U	5	Definitive additional group
T110-30-SC	110	30	SC	5	Definitive additional group
T110-60-U	110	60	U	5	Definitive additional group
T110-60-SC	110	60	SC	5	Definitive additional group
T110-90-U	110	90	U	5	Definitive additional group
T110-90-SC	110	90	SC	5	Definitive additional group

Note: SC denotes the support selected after the screen. AP is not assigned a nominal oven temperature or duration. Total definitive records = 95.

Table A2. Independent confirmation matrix.

Group code	Test	Condition	n
C-U-OPT	Tensile/dimensional	Thermally matched U at model-selected temperature and duration	5
C-SC-OPT	Tensile/dimensional	Selected support at model-selected temperature and duration	5
F-AP	Flexural	As-printed baseline	5
F-U-OPT	Flexural	Thermally matched U at model-selected condition	5
F-SC-OPT	Flexural	Selected support at model-selected condition	5

Note: OPT is a symbolic code assigned only after model selection; it is not a claimed numerical optimum. These 25 specimens are new and are not reused from the screen or factorial.

Appendix B. Specimen Identification Scheme

Identifiers use STAGE-CONDITION-TEMPERATURE-DURATION-REPLICATE. Stage codes are S (screen), D (definitive), C (tensile confirmation) and F (flexural confirmation). Condition codes are AP, U, SAND, SALT, FIX and SC. Temperatures use T080, T095 or T110; durations use M030, M060 or M090; and replicates use R01-R05. AP identifiers omit thermal fields. Examples are S-SAND-T095-M060-R02 and D-U-T080-M030-R04. The final support name remains in the original screen ID, while SC is the analysis alias used for the selected method in the paired factorial.

Specimen ID alone does not establish independence. Separate fields retain print plate, oven run, positions, operator and machine-data filename. Labels stay outside the gauge section without notching. Replaced IDs are retired with the failure reason and are never recycled.